LAB EXPERIMENT RECORD

20-10-2026---Set 22: Time Series Analysis

Department	AIML
Course Code / Course Name	DSA0601 / Data Handling and Visualization
Experiment	Data Visualization using R and Tableau
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Day	Tuesday

Aim

To analyze monthly sales data in R and Tableau using a time series line chart, scatter plot, autocorrelation plot and an interactive dashboard.

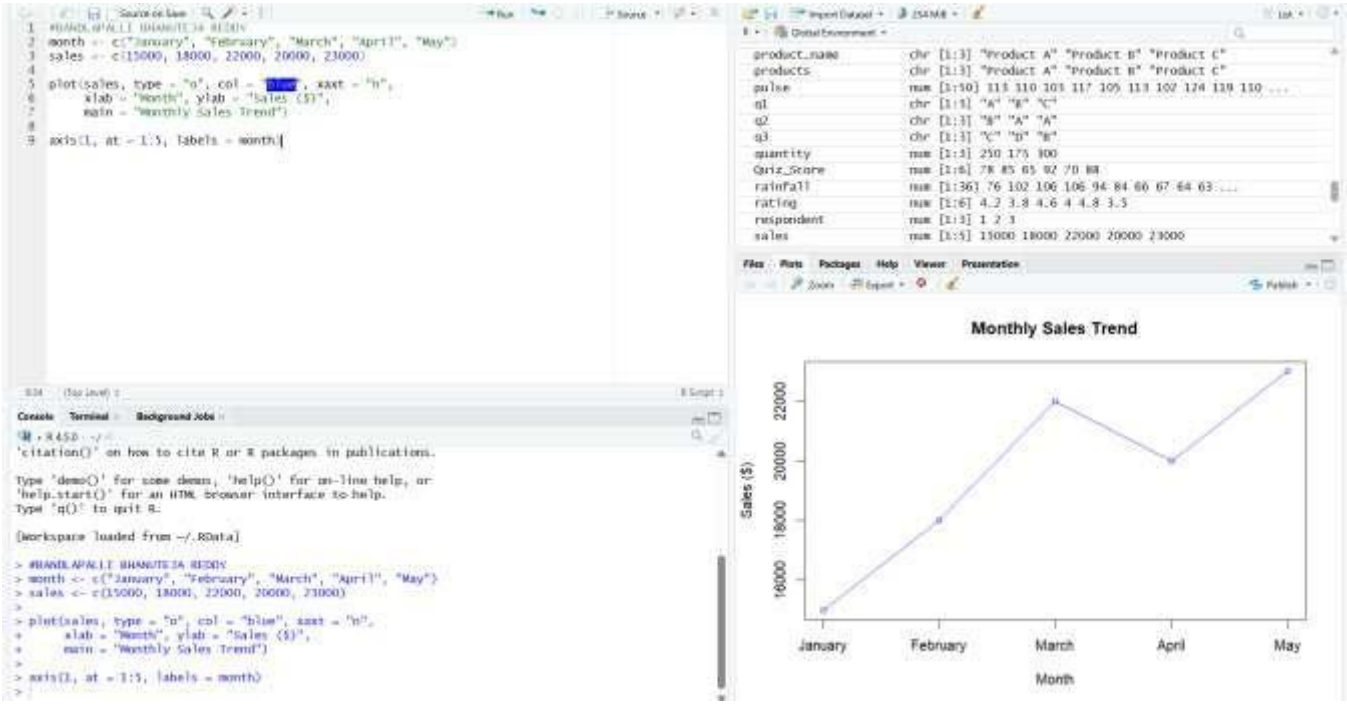
Dataset: Monthly Sales Data

Month	Sales (in $)
January	15000
February	18000
March	22000
April	20000
May	23000

Question 1: Time Series Line Chart of Monthly Sales

Question: In R, create a time series line chart to visualize the trend in monthly sales. Label the axes and title the chart.

Aim: To visualize the trend in monthly sales over time using a time series line chart with labelled axes and a title.

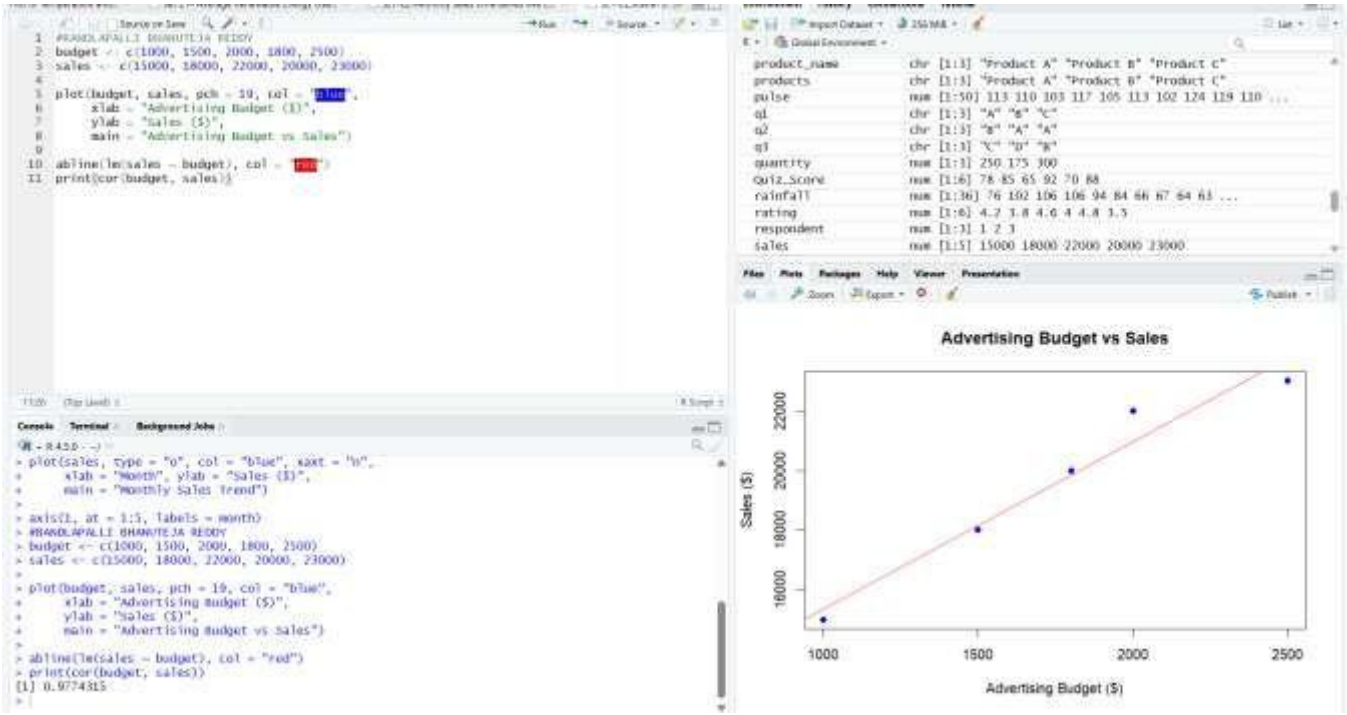


Result: Sales rise from 15000 in January to 22000 in March, dip to 20000 in April and reach the highest value of 23000 in May, so the overall trend is upward.

Question 2: Scatter Plot of Advertising Budget vs Monthly Sales

Question: In R, generate a scatter plot to analyze the relationship between advertising budget and monthly sales. Explain any insights.

Aim: To analyze how advertising budget and monthly sales are related using a scatter plot.

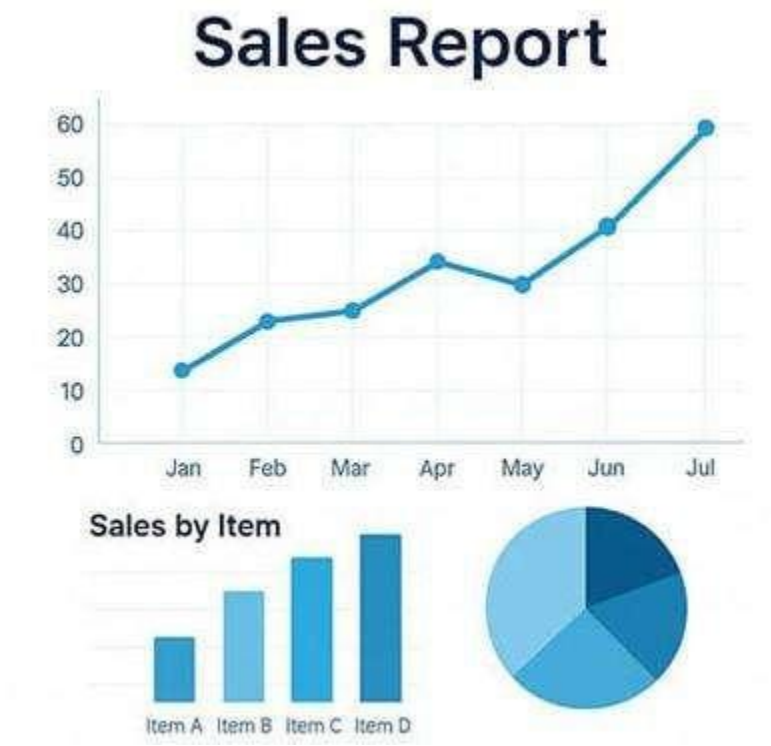


Result: The plot shows whether months with a higher advertising budget have higher sales; the dataset has no advertising budget column, so a conclusion needs those values.

Question 3: Tableau Dashboard of Sales Trend and Product Distribution

Question: In Tableau, build a dashboard combining a line chart showing sales trend and a pie chart displaying the distribution of sales across products.

Aim: To build an interactive Tableau dashboard that combines a sales trend line chart and a pie chart of sales by product.

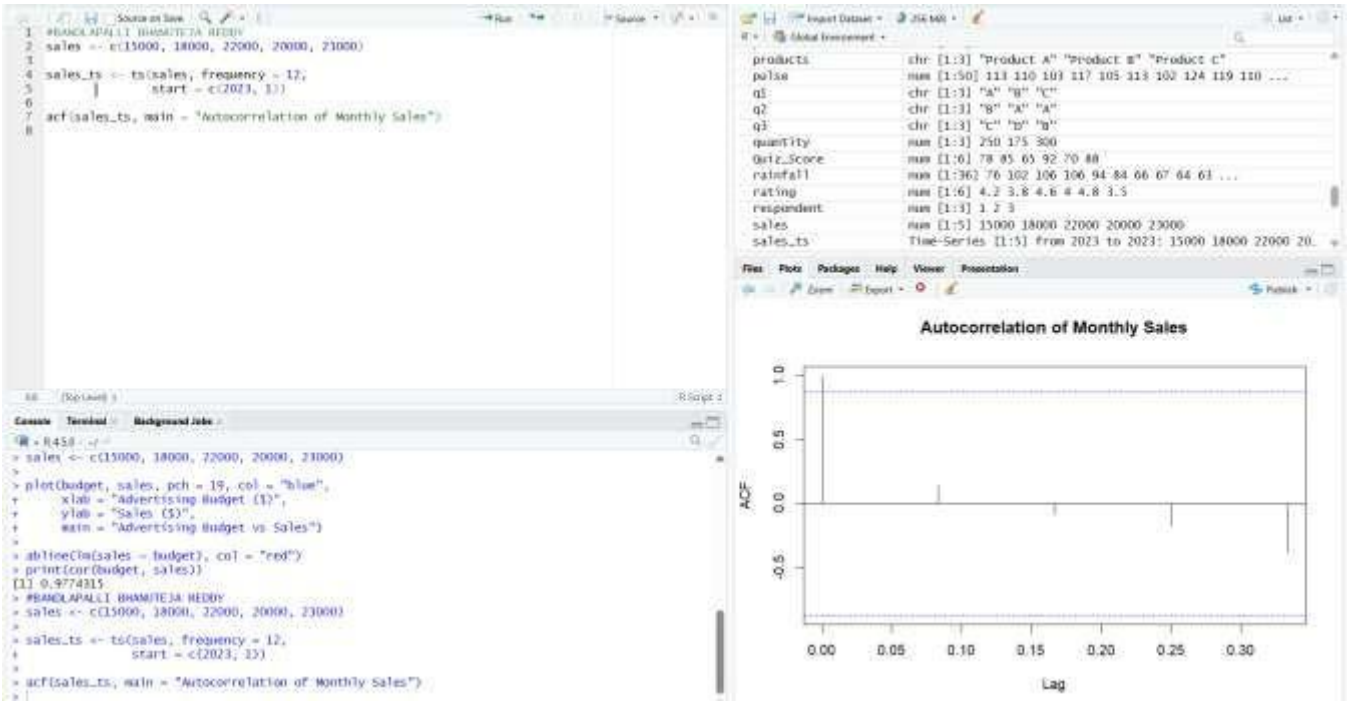


Result: The dashboard places the trend line and the pie chart together; the dataset has no product column, so product-wise sales must be supplied for the pie chart.

Question 4: Autocorrelation Plot for Seasonality

Question: In R, create an autocorrelation plot to identify seasonality in the time series data.

Aim: To check for repeating seasonal patterns in the sales using an autocorrelation plot.



Result: The autocorrelation at lag 1 is only about 0.14 and all lags stay inside the confidence bands, so no seasonality is visible; five months are too few to judge, and at least two full years of data are needed.

Overall Result

Observation: Total sales over the five months are 98,000 with an average of 19,600 per month, and May has the highest sales (23,000).